

COURSE: CHE 312: Transport Phenomena II**Fall Semester 2025****Department of Chemical Engineering****University of Illinois Chicago****Quiz - Exam****Duration: 70 minutes --- October 23, 2025****Both Problems must be done****Problem 1: (50 points)**

Let's design a shell and tube heat exchanger to heat a stream with $m_c C_{p,c} = 5,000 \text{ BTU} / \text{h}^\circ\text{F}$ from 90°F to 210°F flowing in the tubes using as the heat source a stream with $m_h C_{p,h} = 10,000 \text{ BTU} / \text{h}^\circ\text{F}$ entering the shell at 260°F . The heat exchanger will contain tubes with a 0.5 in. outside diameter. In order to maintain a reasonable fluid velocity inside the tubes, $N_t = 30$ tubes per pass will be used. The overall heat transfer coefficient is $U_o = 150 \text{ BTU} / \text{h ft}^2 \text{ }^\circ\text{F}$. For a heat exchanger with one-tube pass and one-shell pass, determine the required tube length. **(40 points)**

Suppose that we need to have shorter tubes, what can we do? **(10 points)**

Heat duty & heat outlet

120

We have

$$m C_{p,c} = 5000$$

$$T_{c,out} = 210$$

$$T_{c,in} = 90$$

Energy balance on hot side

$$Q = m C_p (T_{h,in} - T_{h,out}) \Rightarrow 10000 (260 - T_{h,out}) = 6 \times 10^5 \text{ BTU/h}$$

$$T_{h,out} = 200^\circ\text{F}$$

Temperature driving force

$$\Delta T_1 = T_{h,in} - T_{c,out} = 260 - 210 = 50^\circ\text{F}$$

$$\Delta T_2 = T_{h,out} - T_{c,in} = 200 - 90 = 110^\circ\text{F}$$

$$\Delta T_{\text{em}} = \frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2 / \Delta T_1)} = \frac{60}{\ln(110/50)} = 76.10^\circ\text{F}$$

Required heat transfer area and tube length

$$A = \frac{Q}{U_o \Delta T_{\text{em}}} = \frac{6 \times 10^5}{150 \times 76.10} = 52.56$$

$$A = N_t \pi d_o L \Rightarrow L = \frac{A}{N_t \pi d_o} = \frac{52.56}{30 \pi (0.04167)}$$

$$= \underline{\underline{13.4 \text{ ft}}}$$

The required tube length = 13.4 ft 50×5

① If shorter tubes are required for the job

we could,

increase the tube count to say 40 (example)

which would change the ~~the~~ required length

$$L = 13.4 \times \frac{30}{40} = \underline{\underline{10 \text{ ft}}}$$

~~UNIT 1 (Control System)~~

Reduce the area by splitting duty across two exchangers in parallel.

Or we could raise the value of U by boosting the velocities.

Problem 2: (60 points)

Let's consider a case where 1 kg of aluminum spheres with radius $R = 10$ mm initially at 90°C are immersed in 1 L of water initially at 20°C in a tank. The tank is insulated and heat losses are negligible. The water is assumed to be perfectly mixed. Determine the time needed to cool the aluminum spheres to 40°C assuming that the external heat transfer resistance at the sphere-fluid interface is dominant and that $h = 500 \text{ W/m}^2\text{K}$. The density of aluminum is $\rho = 2,700 \text{ kg/m}^3$ and its heat capacity is $C_p = 900 \text{ J/kg K}$. Assume that the water temperature remains constant. (30 points)

If the changing temperature of the water is considered, determine the time needed to cool the aluminum spheres to 40°C . (30 points)

$$Bi = \frac{hR}{K_{Al}} \ll 0.1 \quad \left(\begin{array}{l} \text{resistance at the sphere-fluid} \\ \text{interface is dominant} \end{array} \right)$$

Case 1 Water temperature constant

Energy balance on the sphere

$$m_s C_p \frac{dT_s}{dt} = -hA_s (T_s - T_\infty)$$

$$\frac{dT_s}{dt} = \frac{hA_s}{m_s C_p} (T_s - T_\infty)$$

$$\int \frac{dT_s}{(T_s - T_\infty)} = \int \frac{hA_s}{m_s C_p} dt$$

$$\ln\left(\frac{T_s - T_\infty}{T_{s,i} - T_\infty}\right) = -\frac{hA_s}{m_s C_p} t \Rightarrow t = \frac{m_s C_p}{hA_s} \ln\left(\frac{T_{s,i} - T_\infty}{T_{s,f} - T_\infty}\right)$$

$$V_{\text{total}} = \frac{m}{\rho} = 3.70 \times 10^{-4} \text{ m}^3$$

$$\text{Volume of one sphere} = 4\pi R^3 = 4.19 \times 10^{-6} \text{ m}^3 \Rightarrow N = \frac{V_{\text{total}}}{V_i} = 88.3$$

$$A_s = N(4\pi R^2) = 88.3 \times 4\pi (0.01)^2 = 0.111 \text{ m}^2$$

$$\therefore t = \frac{900}{500(0.111)} * \ln\left(\frac{90-20}{40-20}\right) = \frac{900}{55.5} \ln(3.5) \quad 30 \times 5$$

$$= 16.2 \times 1.253 = \boxed{20.3 \text{ s}}$$

Case 2: Water temperature changes (no heat loss)

$$\text{Energy balance: } m_s C_p (T_s - T_{s,i}) + m_w C_w (T_w - T_{w,i}) = 0$$

$$m_s C_p \frac{dT_s}{dt} = -h A_s (T_s - T_w) \rightarrow \text{Lumped model differential eq.}$$

$$m_w C_{pw} \frac{dT_w}{dt} = -h A_s (T_s - T_w)$$

$$(T_s - T_w) \text{ elimination: } \frac{d(T_s - T_w)}{dt} = -h A_s \left(\frac{1}{m_s C_p} + \frac{1}{m_w C_{pw}} \right) (T_s - T_w)$$

$$\text{Let } \theta = T_s - T_w \quad \frac{d\theta}{dt} = \frac{dT_s}{dt} - \frac{dT_w}{dt}$$

$$\frac{d\theta}{dt} = -h A \left(\frac{1}{m_s C_s} + \frac{1}{m_w C_w} \right) \theta$$

$$\frac{d\theta}{\theta} = -h A \left(\frac{1}{m_s C_s} + \frac{1}{m_w C_w} \right) dt$$

$$\text{we get } \ln\left(\frac{\theta(t)}{\theta}\right) = -h A \left(\frac{1}{m_s C_s} + \frac{1}{m_w C_w} \right) t$$

$$\text{or } \theta(t) = \theta_0 e^{-\beta t} \quad \because \beta = h A \left(\frac{1}{m_s C_s} + \frac{1}{m_w C_w} \right)$$

$$\text{Substituting } \theta = T_s - T_w$$

$$(T_s - T_w) = (T_{s,i} - T_{w,i}) e^{-\beta t}$$

$$\beta = 500(0.111) \left(\frac{1}{900} + \frac{1}{4180} \right) = 0.0748 \text{ s}^{-1}$$

$$T_s(t) = T_{eq} + (T_{s,i} - T_{eq}) e^{-\beta t}$$

$$\text{at } 40^\circ\text{C} \Rightarrow 40 - 32.1 (90 - 32.1) e^{-0.0748 t} = \frac{7.9}{57.9} = 0.136$$

$$t = \frac{\ln\left(\frac{1}{0.136}\right)}{0.0748} = \boxed{26.7 \text{ sec}}$$

$$\text{Final steady state at } T_{eq}$$

$$m_s C_p (T_{eq} - T_{s,i}) + m_w C_w (T_{eq} - T_{w,i}) = 0$$

$$T_{eq} = \frac{m_s C_p T_{s,i} + m_w C_{pw} T_{w,i}}{m_s C_p + m_w C_{pw}}$$